Voltage Protector

Design report for Wokwi and Proteus 9

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Design approach

With default settings, Voltage Protector Plus disconnects a simulated appliance when voltage falls below 180 V, rises above 250 V, or current exceeds 5 A. The load starts OFF and reconnects after 3 s of continuously healthy input. Two 5 V potentiometers emulate 0-300 V and 0-10 A. A green LED represents the powered load; a red LED indicates FAULT or LOCK.

Two implementations

Simulator	Controller and switching circuit	UART profile
Wokwi	Arduino Uno, 16x2 LCD and two relay modules.	115200 baud
Proteus 9	ATmega328P at 16 MHz, LM016L and two 5 V relays with 2N2222 drivers and 1N4007 flyback diodes.	9600 baud

Wokwi supports functional checks, while Proteus includes discrete relay drivers. Both designs share one protection algorithm with normally open relay contacts in series. SET, UP and DOWN edit the limits and delay; UART commands provide computer control. ACK requests release from LOCK after the fault is removed. EEPROM retains settings and lockout. Recovery uses 5 V voltage and 0.2 A current margins to avoid switching near a limit.

Three practical additions

Contact feedback. A2 reads the main switched node. A sustained command/contact mismatch latches LOCK. The backup is commanded OFF before the main relay, so it can isolate the load if the main contact stays closed.

Repeated fault handling. Three distinct ON-to-fault trips within a rolling 60 s window latch LOCK. The second recovery waits at least 10 s. One sustained fault counts once; safe ACK starts a fresh recovery delay.

Event history. LOG returns the last eight RAM events with timestamps, readings, active thresholds and cause, helping explain why a trip occurred.

Recorded results

Wokwi. Tests showed normal ON, fault shutdown, first recovery of 3.000 s, second recovery of 10.004 s and third-trip lockout. Main-contact bypass injection kept the load OFF through the backup. Local setting changes, LOG and retained EEPROM state were demonstrated. Across both profiles, 90 host behavior checks and 16 AVR HEX checks passed.

Proteus 9. Screenshots show firmware/LCD operation, local settings screens, OVERCURRENT OFF and OVERVOLTAGE OFF. At 252.1 V and 2.00 A, the over-voltage state showed the red LED ON, green load OFF and relay contacts in their OFF positions.

Scope and remaining verification

These are low-voltage simulations with emulated readings, without AC RMS sensing or a measured physical disconnect time. Only the main contact has feedback. Proteus relay-transition, UART and contact-fault injection tests still need recorded verification. Hardware sensing, relay ratings and response measurements remain future work.

Evidence: Wokwi screenshots and serial logs, host/HEX check logs, and Proteus screenshots dated 10 October 2026.